LIBBY — PATIENT/CAREGIVER TRACK

aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun

Plain-language summary for patient and caregiver

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This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

What this page is

Libby is a research tool that looks at the specific features of one person's cancer and gathers the treatment options built to target those features, along with the trials that offer them. This page is the plain-language version of that work. It walks through what we learned about your leukemia, what you said mattered to you, the one test worth doing before anything else, and then the options themselves, ranked, with the upside and the catch of each laid out as plainly as we can.

It is not a treatment plan and it is not a substitute for your oncologist. Treat it as an organized starting point for the conversation with your care team.

What we know about your cancer

You have acute myeloid leukemia, or AML, that grew out of an earlier bone-marrow disorder (MDS) you were diagnosed with several years ago. The leukemia came back about six years after a stem-cell transplant from a matched sibling donor, which is a long remission. The pattern suggests the leukemia found a way around the donor immune cells rather than the graft simply failing.

A few things about where the disease stands now shape the options below:

- **The marrow is full of leukemia.** Roughly 85% of your bone marrow is leukemia cells right now, and the count in your blood has been climbing. There is also a solid deposit of leukemia in the right thigh, which doctors treat as an aggressive sign.
- **The last chemotherapy did not work.** A strong reinduction regimen was tried and the leukemia kept growing through it, so ordinary chemotherapy alone is unlikely to clear this marrow.
- **The leukemia carries two surface proteins we can aim at.** Lab testing confirmed the cells carry proteins called CD33 and CD123. Both are handles that certain targeted drugs can grab. What is not yet known is how much of each protein the cells carry, and the amount matters for how well those drugs work.
- **The tissue-type markers that the main plan hinges on have not been pulled yet.** The immunotherapy pathway you asked us to look at depends on a tissue-type marker called HLA-A*02:01, and that result is not in the records on hand. More on this just below, because it is the first step.
- **A gene called TP53 is unsettled.** One test years ago flagged a TP53 change; a more recent, more limited test did not find it. This has not been resolved either way, and it is not being treated as settled. If a specific TP53 change is confirmed later, it could open a different drug avenue that is not on this page yet.
- **The chromosome picture is complex.** The leukemia cells carry several chromosome changes that put it in a higher-risk group. This is part of why the options lean toward a transplant-based, curative plan rather than a milder one.

You are up and about and able to care for yourself day to day. One gap worth naming: there are no recent heart, kidney, or liver lab numbers on file, and several options below cannot safely start until that fitness check is done.

What you told us matters most

- **You want a shot at a cure.** The plan you asked for is aimed at long-term control, not just holding the line, and the ranking reflects that.
- **You lean toward the strongest treatment, and you accept more side effects to get it.** That lean is set

provisionally. It should be revisited once the fitness check comes back, because how much treatment your body can safely take is still an open question.

- **You are open to clinical trials, including travel for them.** Several of the most fitting options live inside trials, so trials are surfaced wherever they fit rather than treated as a last resort.
- **A second transplant and cell therapy are both on the table for you.** You did not rule out any type of treatment, which matters because the leading plan below is built around a second transplant.
- **The specific reason you came to Libby: the immunotherapy pathway.** You asked us to look hard at a form of engineered-immune-cell therapy that could work after a transplant. Surfacing that pathway and figuring out whether you are eligible is the priority, and it is exactly what the first step below is designed to answer.

The first step everyone agreed on

Before any of the immunotherapy options can even be scoped, one set of tests has to come back. This is the single point of full agreement in the whole analysis, and it is not a drug.

What it is: high-resolution tissue typing to pin down your HLA-A*02:01 status (and a second marker, HLA-A*24:02, checked at the same time), plus a genetic test called HA-1/HA-2 genotyping done on both you and your sibling donor. HA-1 and HA-2 are tiny cell-surface differences that even a well-matched sibling can differ on, and the immunotherapy below only works if you and your donor differ in a specific way.

Why it comes first: HLA-A*02:01 is the master switch for the engineered-immune-cell therapy you asked about. If it is present, that option is on the table. If it is absent, that option is off the table. There is no in-between, and nothing about the immunotherapy path can be decided until this result is in hand.

What it costs you: essentially nothing beyond time. The tissue typing is a blood draw or, better, a record that may already exist from your first transplant, and it turns around in a few days. The HA-1/HA-2 test is a more specialized send-out that takes about one to three weeks and needs a fresh sample from your sibling donor, so it is worth getting your donor's consent lined up early. Neither test has side effects.

There are really three things this step is trying to answer: whether HLA-A*02:01 is present, whether you and your donor carry the specific HA-1 difference the therapy needs, and whether the old typing record can be retrieved so you do not have to repeat the draw.

The options

Six treatment options came out of the analysis, in the order below. The workup above is the true first step; the numbering here covers the therapies that follow it. The first two are the core of the plan. Options 3 through 6 are on the page because Libby shows what was weighed, including what was set aside, and each carries a specific caution spelled out plainly.

Option 1 — Targeted-radiation conditioning (Iomab-B) followed by a second transplant

This is the backbone of a curative plan, and it is the one option everyone in the analysis endorsed without reservation. The idea is to use an antibody that carries a dose of radiation directly to the leukemia in your marrow, clearing it out, and then to give you a second stem-cell transplant. What makes this approach different from ordinary pre-transplant chemotherapy is that it is designed to carry active leukemia into the transplant rather than demanding that chemotherapy get you into remission first, which chemotherapy has not been able to do here.

In the main trial, which enrolled 153 people, about 17 out of every 100 who got this targeted-radiation conditioning were in a deep, lasting remission six months later, compared with none of the people on standard care. People on this approach were also far more likely to actually reach a transplant and stay free of relapse. One honest limit: across everyone in the trial, people did not, on average, live longer, partly because most of the standard-care group crossed over to receive this same treatment once their leukemia progressed, which muddies that particular comparison.

There is a second, bigger caution to weigh. That trial enrolled people heading to their first transplant. You would be heading to a second, and the risk of a second transplant in someone who has already had one is not something the trial measured. So the remission numbers above are a fair expectation for what the conditioning does, not a proven figure for your exact situation.

On side effects, the honest read is that this conditioning adds very little on top of what any transplant already involves: serious side effects happened in about 60 out of 100 people in both the treatment and the standard-care groups, so the radiation antibody is not the driver. The usual transplant burdens still apply: very low blood counts, infections, and mouth sores. And this cannot start until the heart, kidney, and liver fitness check is done, because that clearance is a hard prerequisite.

This fits your curative goal and your openness to a second transplant. It is also the platform the immunotherapy in Option 2 would be delivered on top of, which is part of why it sits so high.

Option 2 — Engineered immune-cell therapy targeting HA-1 (TCR-T)

This option is only available if the HLA-A*02:01 test comes back positive, together with the specific HA-1 difference between you and your donor. If HLA-A*02:01 comes back negative, this option is off the table. And because this page only ranks options that target your specific handles, a negative result means there are no other options ranked here on this immunotherapy path. There is one narrow exception: the second marker checked at the same time (HLA-A*24:02), if it is positive, could open a different engineered-immunotherapy avenue, but that is something your transplant team would pursue directly. Standard care for relapsed AML exists and is worth discussing, but it sits outside what this page ranks; that is a separate conversation with your oncologist.

With that said, this is the option that answers the question you came to Libby with. It takes immune cells and engineers them to recognize HA-1, a marker that sits only on blood-lineage cells, so the engineered cells hunt the leukemia while mostly leaving the rest of the body alone. The plan would be to give it after the transplant in Option 1, as a way to keep the donor immune system attacking any leukemia that remains.

What the evidence looks like: in an early study of 9 people who had relapsed after a transplant, 4 reached or held a complete remission, and one of them stayed in remission beyond two years. Just as striking, there were no cases of the severe immune overreaction (cytokine release) or the neurological side effects that this kind of cell therapy can cause, across all 9 people. The main side effects came from the chemotherapy given to make room for the cells: low blood counts and serious infections, plus two mild cases of graft-versus-host reaction.

Here is the open question to hold onto, and it does not go away even if your test comes back positive: this is 9 people in an early study where everyone got the treatment, with no comparison group. That means no one can cleanly separate how much of the benefit came from the engineered cells versus the chemotherapy or the natural course of the disease. The safety signal is genuinely clean; the effectiveness is promising but unproven. One more practical point: with the marrow this full of leukemia, this therapy is not a standalone. It needs the leukemia knocked down first and the transplant platform in place, so it comes late in the sequence, not first.

Option 3 — Tagraxofusp (a CD123-targeted drug)

This option carries caveats. Here is the tradeoff to weigh.

Tagraxofusp aims at CD123, one of the two proteins your leukemia carries. It is a drug that uses the CD123 handle to deliver a toxin into the cancer cell.

The reason it is here rather than higher: the eye-catching number attached to it, a response in about 72 out of 100 people, comes from a different blood cancer (a rare one called BPDCN), not from AML. There is no AML effectiveness data for it, so that number does not carry over to a marrow as full of leukemia as yours. It targets a protein you genuinely have, which is the argument for it, but the amount of CD123 on your cells, which predicts how well it would work, has not been measured yet. A confirmed high amount would be a stronger signal than what we can say today, so this stays an uncertain option.

The side effects are the other reason for caution. This drug can cause capillary leak syndrome, where fluid leaks out of small blood vessels, and there were deaths from it in the study. It also strains the liver in most people who take it. With no baseline liver, heart, or blood-protein labs on file, this is exactly the kind of drug that needs a fitness check before it could safely start. The one thing in its favor on safety is that, because it has been used for years, doctors have a worked-out routine for watching and managing the capillary-leak risk.

Option 4 — Gemtuzumab ozogamicin (a CD33-targeted drug)

This option carries caveats. Here is the tradeoff to weigh.

This drug aims at CD33, the other protein your leukemia carries. It is an antibody that carries a toxin straight to CD33-positive cells, and unlike most of this list, it is fully approved and guideline-listed for relapsed CD33-positive AML. On the surface, that makes it the familiar, well-covered choice.

The catch is specific and important. The strongest evidence for this drug, drawn from more than 3,000 patients, found no survival benefit in the group whose chromosomes look like yours, the complex, higher-risk group. In other words, the best data available point away from it helping in your particular situation, not toward it. Having the CD33 target present and having the drug actually help are not the same thing here.

There is also a side-effect reason to be careful: this drug carries a risk of a serious liver complication called veno-occlusive disease, which is the wrong risk to invite right before a second transplant. Like the CD123 drug above, the amount of CD33 on your cells (and a genetic detail that also predicts response) has not been measured, so even the case for it rests on unknowns. It stays on the page as the on-label option against a target you carry, but its own best evidence is pointed the wrong way for you.

Option 5 — Pivekimab sunirine (a CD123-targeted drug)

This option was considered but not recommended. It is on the page because Libby shows what was weighed and set aside, not only what was kept. Here is what the concern was.

Pivekimab also aims at CD123, and of the CD123 drugs here, it has the best data in AML specifically, rather than borrowed from another cancer: a response in about 21 out of 100 people with relapsed AML. The thought was to use it as a way to knock the leukemia down before the transplant and cell therapy.

It is held back for one concrete safety reason. At higher doses it caused veno-occlusive disease, that same serious liver complication, and there was one treatment-related death in the study. Bringing a drug with a known liver-clotting risk right up against a planned transplant, in someone with no liver labs on file, is the specific worry. This is a conditional hold, not a permanent no: if baseline liver and blood-protein labs come back clean, this reopens as a reasonable option for shrinking the leukemia before transplant. It is kept visible precisely so you can see what it would take to put it back in play, rather than having it quietly dropped.

Option 6 — Flotetuzumab (a CD123-and-T-cell drug)

This option carries caveats. Here is the tradeoff to weigh.

Flotetuzumab is a drug designed to grab CD123 on the leukemia with one end and a T cell with the other, pulling the immune system directly onto the cancer. In a study of relapsed AML, it produced a response in about 30 out of 100 people, and there was an interesting observation that some patients with TP53-altered leukemia responded.

Two honest limits keep this near the bottom. First, the drug itself is no longer being developed and there is no open trial to enroll in, so it is not actually a route you could take right now. Second, the TP53 observation, the part that might seem to fit your situation, came from a small after-the-fact look at 7 out of 15 patients, which makes it a hypothesis rather than a measured result. The real value here is the mechanism: it shows that pulling T cells onto CD123 can work as a salvage approach, which is a reason to look for an open trial testing a similar newer drug, especially if your TP53 status is confirmed. The main side effect of this class is an immune reaction during infusions, usually mild and managed with step-up dosing and standard medicines.

Questions to ask your oncologist

- Can we get the HLA-A*02:01 and HLA-A*24:02 tissue typing and the HA-1/HA-2 genotyping started now, including retrieving the old typing record and lining up my sibling donor's sample, and how soon will each come back?
- If the HLA-A*02:01 test comes back negative, what does that leave us with, and would the second marker (A*24:02) open any alternative immunotherapy path worth checking?
- For the targeted-radiation conditioning and second transplant, what fitness tests do I need first (heart, kidney, liver), and is there an open trial that accepts someone who has already had a transplant?
- Given that a second transplant is riskier than a first and that risk was not measured in the trial, how would you counsel me on that unknown?
- The engineered immune-cell therapy needs my leukemia knocked down and a transplant in place first. What is the plan and timeline for getting there while the marrow is this full?
- Can we measure how much CD33 and CD123 my leukemia cells actually carry, since that predicts whether the CD33- and CD123-targeted drugs would help?
- For the CD123 drug that was held back over a liver-clotting risk (pivekimab), what liver and blood-protein labs would you need to see to consider it as a way to shrink the leukemia before transplant?
- Is my TP53 status worth resolving now, and if a specific TP53 change is confirmed, does that change my options?

Sources

PubMed (PMID): 11432892, 22482940, 25008258, 29661755, 31018069, 31275297, 32929488, 33057635, 38423051, 38683966, 39298738

ClinicalTrials.gov (NCT): NCT02152956, NCT02665065, NCT03326921, NCT03386513, NCT05473910, NCT06561152, NCT07157514